

Dynamic Guideline: WHO-Based Fascioliasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO fascioliasis-related guideline document

- **Guideline Title:** *Fasciola: background document for the WHO guidelines for drinking-water quality and the WHO guidelines on sanitation and health*
- **Publication Date:** 1 March 2025
- **Publisher:** World Health Organization
- **Document Type:** WHO guideline-related background document
- **Target Population:** Populations exposed to *Fasciola* transmission through unsafe drinking-water, contaminated surface water, poor sanitation, animal faecal contamination, and ingestion of contaminated aquatic or semi-aquatic plants.
- **Scope and Objectives:** To support WHO drinking-water and sanitation guidance by summarizing evidence on *Fasciola* as a water-, sanitation-, zoonosis-, and food-linked helminth hazard. WHO reports **moderate evidence** linking *Fasciola* transmission with **unsafe drinking-water and inadequate sanitation**, and notes that ingestion of contaminated surface water is supported by indirect epidemiologic evidence [WHO Fasciola background document](#).

Exactly 5 WHO / WHO-linked guidelines or guidance documents synthesized

No.	WHO / WHO-linked guideline or guidance document	Date identified in available sources	Relevance to fascioliasis	Source
1	<i>Fasciola: background document for the WHO guidelines for drinking-water quality and the WHO guidelines on sanitation and health</i>	1 March 2025	Latest WHO fascioliasis-specific water/sanitation risk-management document; identifies moderate evidence linking transmission with unsafe water and inadequate sanitation.	WHO

No.	WHO / WHO-linked guideline or guidance document	Date identified in available sources	Relevance to fascioliasis	Source
2	<i>Operational guidelines for the elimination of human fascioliasis as a public health problem in the Americas</i>	Date not visible in supplied summary; active for 2030 elimination target	WHO/PAHO operational guidance for elimination; defines elimination target as sustained prevalence $\leq 5\%$ by coprology and no high-intensity infections ≥ 400 eggs/g faeces.	WHO publication page, PAHO/WHO PDF
3	<i>WHO Guidelines for drinking-water quality: Fourth edition incorporating the first and second addenda</i>	2022	Includes <i>Fasciola</i> spp. among helminth pathogens and frames drinking-water guidance around public-health risk management.	WHO drinking-water quality PDF, NCBI Bookshelf
4	<i>WHO Guidelines on sanitation and health</i>	Referenced as parent guideline by 2025 WHO <i>Fasciola</i> document	Relevant because <i>Fasciola</i> transmission is linked to faecal contamination, animal faeces management, unsafe water, and sanitation service-chain control.	WHO Fasciola background document
5	WHO foodborne trematode infection control guidance / recommendations for community diagnosis and preventive chemotherapy	Active WHO guidance as of 29 April 2026	Identifies fascioliasis as a major foodborne trematode infection and recommends triclabendazole as the only WHO-recommended medicine for fascioliasis.	WHO recommendations, WHO activity page, WHO fact sheet

Methodology note on the requested WHO youth-engagement title: The most recent WHO item found with the title pattern “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” is a cross-cutting WHO evidence brief on guideline-development methodology, not a fascioliasis guideline. WHO scheduled a launch webinar for this brief on **11 December 2025**. The available evidence does **not** show that meaningful youth engagement was incorporated into the WHO/PAHO fascioliasis guideline process [WHO publication, WHO webinar](#).

1.2 Key Recommendations

Recommendation	Practical action	GRADE Evidence Quality	Strength
Treat confirmed or strongly suspected human fascioliasis with triclabendazole .	WHO recommends triclabendazole 10 mg/kg orally as a single dose ; in treatment failure, increase to 20 mg/kg total , given as two divided doses 12–24 hours apart. Give with food where feasible to improve absorption.	Moderate	Strong
Use triclabendazole as first-line therapy because it is active against both immature and adult flukes.	Use in acute and chronic disease when clinical, epidemiologic, serologic, imaging, or parasitologic evidence supports fascioliasis.	Moderate	Strong
Do not rely on praziquantel for fascioliasis.	If a patient was empirically treated with praziquantel and eosinophilia/symptoms persist, reassess for fascioliasis and use triclabendazole when indicated.	Moderate	Strong
Diagnose chronic fascioliasis by detecting <i>Fasciola</i> eggs in stool, duodenal aspirate, biliary drainage, or other appropriate specimens when possible.	Use repeated stool microscopy, rapid sedimentation, Kato-Katz, or duodenal/biliary sampling where available.	Moderate	Strong
Use serology, eosinophilia, imaging, exposure history, and clinical syndrome to support diagnosis, especially in acute disease when eggs may be absent.	Consider serologic tests such as CDC's FhSAP2 immunoblot in appropriate settings; interpret in clinical context.	Moderate	Conditional
Implement water, sanitation, hygiene, and food-safety interventions in endemic areas.	Prevent ingestion of contaminated raw aquatic/semi-aquatic plants and untreated surface water; improve sanitation and animal faeces management.	Moderate	Strong
Manage <i>Fasciola</i> as a One Health problem.	Coordinate human health, veterinary, water, agriculture, and sanitation sectors; reduce livestock-driven contamination and avoid unnecessary triclabendazole use in animals.	Low to Moderate	Strong
Monitor treatment efficacy where failures occur or transmission is intense.	Use post-treatment stool microscopy, fecal egg count reduction, or coproantigen testing where available; distinguish reinfection from true failure.	Low	Conditional

Recommendation	Practical action	GRADE Evidence Quality	Strength
Use preventive chemotherapy only where epidemiology supports clustered or high-burden infection.	Consider population-based treatment in endemic communities with high prevalence; individual case management may be sufficient for sporadic cases.	Low to Moderate	Conditional
For drinking-water risk management, do not assume that <i>E. coli</i> or thermotolerant coliforms reliably indicate absence of <i>Fasciola</i> .	Apply water-safety planning and physical removal/barrier approaches where <i>Fasciola</i> risk exists; WHO notes metacercariae are likely resistant to chlorination.	Low to Moderate	Conditional

Key supporting sources: WHO states that triclabendazole is the **only medicine recommended by WHO** for fascioliasis, active against immature and adult parasites, with a standard **10 mg/kg single dose** and a **20 mg/kg divided-dose** option for treatment failure [WHO recommendations](#). WHO's Q&A identifies triclabendazole as first-line treatment and notes rare need for surgery for complications such as bile duct obstruction [WHO Q&A](#). CDC similarly identifies triclabendazole as the drug of choice and recommends two 10 mg/kg doses 12 hours apart for patients ≥6 years in the United States [CDC Clinical Overview](#).

1.3 Guideline Development Methodology

Evidence Review Process

- The latest WHO *Fasciola* background document summarizes pathogen characteristics, transmission pathways, occurrence in water and sanitation systems, detection methods, environmental persistence, and risk-management implications for drinking-water and sanitation guidance [WHO Fasciola background document](#).
- WHO reports **moderate evidence** linking *Fasciola* transmission with unsafe drinking-water and inadequate sanitation, including indirect evidence from statistically significant associations between fascioliasis and other waterborne parasitic infections [WHO Fasciola background document](#).
- The WHO/PAHO operational guideline states that it is based on available technical evidence and, where evidence gaps exist, input from regional experts, the WHO Collaborating Centre on fascioliasis and its vector snails, and national programme managers [PAHO/WHO operational guidelines PDF](#).
- The provided sources did **not** identify a formal fascioliasis-specific WHO GRADE evidence-to-decision table. Therefore, the GRADE ratings in this report are applied transparently using standard GRADE principles: certainty is rated higher for consistent clinical trial/diagnostic evidence and lower for observational, indirect, mechanistic, or geographically limited evidence.

Consensus Method

- WHO/PAHO operational guidance was prepared with support from country technicians, regional experts, and the WHO Collaborating Centre on fascioliasis and vector snails, aiming to incorporate current practices and challenges in endemic areas [WHO publication page](#).

- The 2025 WHO *Fasciola* background document supports broader WHO drinking-water and sanitation guideline frameworks rather than serving as a stand-alone clinical treatment guideline [WHO Fasciola background document](#).

Conflict of Interest Management

- The accessible summaries did **not** provide a fascioliasis-specific WHO author conflict-of-interest declaration.
- WHO drinking-water guidance does include an institutional conflict-of-interest principle: drinking-water surveillance and supplier quality-control functions are best performed by separate, independent entities because combining them creates a conflict of interest [WHO drinking-water guidance copy](#).
- Non-guideline clinical/review publications identified in the evidence summaries reported no conflicts of interest, but these statements do not substitute for WHO guideline-panel conflict management [Semantic Scholar PDF](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

Latest WHO guideline release used as anchor: *Fasciola: background document for the WHO guidelines for drinking-water quality and the WHO guidelines on sanitation and health*, published **1 March 2025** [WHO Fasciola background document](#).

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from WHO, PAHO/WHO, CDC, FDA, NIH/NLM PubMed Central, peer-reviewed journal sources, and clinical-trial registry infrastructure identified in the supplied summaries. Search concepts included:

- *Fasciola*, fascioliasis, diagnosis, stool microscopy, serology, coproantigen testing
- triclabendazole, treatment failure, resistance, dosing
- drinking-water, sanitation, contaminated surface water, foodborne trematodes
- ClinicalTrials.gov, WHO ICTRP, 2025–2026 trials
- WHO guideline development, GRADE, conflict of interest, meaningful youth engagement

Databases and Sources Searched

Source type	Sources identified	Use in this report
WHO official guidance	WHO <i>Fasciola</i> background document; WHO drinking-water quality guidelines; WHO foodborne trematode guidance; WHO Q&A; WHO activity page	Core WHO recommendations, transmission, WASH, treatment access
WHO regional guidance	PAHO/WHO operational guidelines for elimination of human fascioliasis	Elimination targets, One Health, resistance stewardship

Source type	Sources identified	Use in this report
Public-health agencies	CDC clinical overview, CDC DPDx, CDC Emerging Infectious Diseases	Diagnosis, U.S. dosing, serology, resistance/treatment failure
Regulatory evidence	FDA multidisciplinary review and label	Clinical trial evidence supporting triclabendazole approval
Peer-reviewed literature	Frontiers 2025, Nature Communications 2025, PMC reviews/case reports, MDPI 2026	Treatment outcomes, microbiome, resistance genomics, veterinary resistance
Trial registries	ClinicalTrials.gov, WHO ICTRP	Assessment of active/new 2025–2026 trials

Inclusion Criteria

- Human fascioliasis diagnosis, treatment, management, prevention, or control.
- WHO or WHO-linked guidance relevant to fascioliasis, drinking-water, sanitation, foodborne trematodes, or elimination.
- Post-**1 March 2025** evidence where available, especially 2025–2026 literature.
- Authoritative pre-2025 evidence retained only where needed to explain existing WHO/CDC/FDA recommendations.

Exclusion Criteria

- Animal-only studies without One Health or human treatment relevance, except where resistance affects human treatment policy.
- Non-authoritative web content.
- Trial registry infrastructure pages without disease-specific trial records, except to document absence of identified 2025–2026 triclabendazole fascioliasis trials.

2.2 Key New Studies

Study / Source	Design	Key Findings	GRADE Quality
WHO <i>Fasciola</i> background document, 2025	WHO guideline-related evidence review/background document	Reports moderate evidence linking <i>Fasciola</i> transmission with unsafe drinking-water and inadequate sanitation; notes contaminated surface water transmission is supported by indirect evidence.	Moderate
WHO Q&A on fascioliasis, 2025	WHO clinical/public-health summary	States fascioliasis is often misdiagnosed because symptoms resemble hepatitis/cholangitis; diagnosis may use stool egg examination and serology, but access is limited in endemic areas.	Low to Moderate

Study / Source	Design	Key Findings	GRADE Quality
WHO Q&A on fascioliasis, 2025	WHO clinical/public-health summary	Identifies triclabendazole as first-line treatment effective against immature and adult flukes; surgery may rarely be needed for bile duct obstruction.	Moderate
Frontiers in Cellular and Infection Microbiology, 2025	Observational cohort / treatment-outcome microbiome study	In 30 chronic <i>F. hepatica</i> cases treated with two directly observed triclabendazole doses, 13 became stool-negative, 11 remained stool-positive, and 6 lacked follow-up samples.	Low
Nature Communications, 2025	Genomic resistance study	Reports independent origins and non-parallel selection signatures of triclabendazole resistance in <i>F. hepatica</i> ; supports need for molecular resistance diagnostics and stewardship.	Low to Moderate
CDC Emerging Infectious Diseases, 2021, retained for resistance context	Observational pediatric cohort in Cusco, Peru	Reported low triclabendazole efficacy in children with chronic infection: 55% after one round, falling to 38%, 30%, and 23% after subsequent rounds; 11% failed >4 rounds.	Low
PAHO/WHO operational guidance	Regional operational guideline	Recommends treatment-efficacy monitoring using fecal egg count reduction or coproantigen testing where available; warns not to confuse reinfection with resistance.	Low to Moderate
CDC Clinical Overview / DPDx	Public-health clinical guidance	CDC identifies triclabendazole as drug of choice; for patients ≥ 6 years, recommends two 10 mg/kg oral doses 12 hours apart with food; CDC DPDx describes serologic immunoblot with chronic infection sensitivity $\geq 94\%$ and specificity $\geq 98\%$.	Moderate
FDA label / multidisciplinary review	Regulatory review of clinical trial and open-label data	FDA-approved triclabendazole for fascioliasis in patients ≥ 6 years; label reports highest Day 60 cure rate for 20 mg/kg in two divided doses.	Moderate
ClinicalTrials.gov / WHO ICTRP search evidence	Trial-registry review	No specific 2025–2026 ClinicalTrials.gov or WHO ICTRP triclabendazole fascioliasis trial record was identified in supplied results.	Very Low

Selected source links: WHO 2025 document [WHO](#); CDC clinical overview [CDC](#); CDC DPDx [CDC DPDx](#); Frontiers 2025 [Frontiers](#); Nature Communications 2025 [Nature](#); FDA label [FDA](#); CDC EID Peru [CDC EID](#); WHO ICTRP [WHO ICTRP](#).

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Research need	GRADE certainty of current evidence
No recent human systematic review/meta-analysis quantifying triclabendazole failure rates was identified for 2025–2026.	Treatment-failure estimates are geographically heterogeneous and may reflect reinfection, high burden, poor absorption, adherence, or true resistance.	Multicountry pooled analyses using standardized cure and egg-reduction definitions.	Low
Limited validated diagnostics for acute fascioliasis in endemic field settings.	Stool eggs are often absent in acute infection; serology may be unavailable or not validated for acute disease in some settings.	Affordable point-of-care antigen/antibody and molecular tests validated by disease phase.	Low to Moderate
No widely available molecular resistance test for human clinical practice.	Genomic evidence suggests resistance may arise through multiple independent pathways, complicating surveillance.	Field-ready molecular markers and phenotypic-genotypic correlation studies.	Low
Unclear optimal regimen where triclabendazole failure is common.	WHO standard regimen may underperform in some hotspots; CDC/FDA dosing differs from WHO single-dose guidance.	Randomized trials comparing single-dose, two-dose, repeat-dose, and alternative therapies in high-failure settings.	Low to Moderate
Limited evidence on alternatives to triclabendazole.	Triclabendazole is the only WHO-recommended medicine; resistance threatens control.	Trials of oxfendazole, nitazoxanide, combination regimens, and new fasciolicides.	Very Low to Low
Environmental detection and water-treatment evidence remains indirect.	WHO notes <i>E. coli</i> is not a suitable indicator and metacercariae may resist chlorination.	Quantitative microbial risk assessment, water-treatment removal studies, and field validation of <i>Fasciola</i> water surveillance.	Low

Evidence gap	Why it matters	Research need	GRADE certainty of current evidence
Youth/community engagement in fascioliasis guideline development not documented.	Fascioliasis often affects rural children and adolescents, but youth participation was not evident in available fascioliasis guideline snippets.	Incorporate meaningful youth/community engagement in future endemic-area guideline updates.	Very Low

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Treat fascioliasis with triclabendazole 10 mg/kg single dose; use 20 mg/kg divided dose for failure.	2025–2026 evidence reinforces triclabendazole as central therapy but highlights treatment failure and resistance concerns.	Continue WHO regimen as baseline. In high-failure settings, ensure directly observed treatment with food, confirm diagnosis, repeat stool testing, and consider 20 mg/kg divided dosing or expert consultation.
Use stool microscopy and serology for diagnosis.	CDC DPDx emphasizes stool egg detection mainly in chronic disease and FhSAP2 immunoblot performance in chronic infection; acute disease remains difficult.	Use a multimodal diagnostic approach: exposure history, eosinophilia, liver/biliary imaging, serology, coproantigen where available, and repeated stool microscopy. Do not exclude acute fascioliasis solely because stool eggs are absent.
Implement WASH and food-safety prevention.	WHO 2025 document strengthens the water/sanitation link and notes limitations of standard microbial indicators and chlorination.	Treat fascioliasis as a water-, sanitation-, food-, and animal-faeces-management hazard. Use water-safety planning, sanitation-chain control, safe irrigation, safe washing/cooking of aquatic plants, and livestock faeces management.
Use preventive chemotherapy where large numbers are infected.	No new high-quality trial evidence changed the indication threshold. Resistance concerns argue against indiscriminate use.	Use preventive chemotherapy only after local epidemiologic assessment. Pair with surveillance, efficacy monitoring, health education, WASH, and One Health interventions.
Monitor for treatment failure when suspected.	Human treatment failure in Peru and genomic resistance evidence increase urgency.	In endemic hotspots, implement routine post-treatment assessment using repeated stool microscopy, coproantigen testing, or fecal egg count reduction

Original Recommendation	New Evidence Impact	Updated Recommendation
		where available; distinguish reinfection from resistance.
Coordinate veterinary and human-health control.	PAHO/WHO and resistance literature emphasize animal reservoirs and triclabendazole stewardship.	Avoid unnecessary triclabendazole use in animals where possible; consider rotating non-human fasciolicides and integrating veterinary surveillance with human control programmes.
Rely on standard water-quality indicators for microbial safety.	WHO states <i>E. coli</i> or thermotolerant coliforms are not suitable indicators for possible presence or absence of <i>Fasciola</i> .	In endemic areas, do not use absence of <i>E. coli</i> as proof of absence of <i>Fasciola</i> . Apply source protection, surface-water avoidance/treatment, physical removal barriers, and sanitation controls.

Practical Clinical and Public-Health Algorithm

A. When to suspect fascioliasis

Suspect fascioliasis in patients with compatible symptoms plus exposure risk:

- Consumption of raw aquatic or semi-aquatic plants, or plants irrigated/washed with unsafe water.
- Use of untreated surface water from rivers, ponds, canals, or livestock-contaminated sources.
- Residence in or travel to endemic rural areas.
- Eosinophilia with fever, right upper quadrant pain, hepatomegaly, abnormal liver enzymes, or biliary symptoms.
- Chronic biliary disease, cholangitis-like illness, jaundice, or unexplained liver lesions.

WHO notes that fascioliasis is often misdiagnosed because symptoms overlap with hepatitis and cholangitis [WHO Q&A](#).

B. Diagnostic approach

Disease phase	Preferred diagnostic tools	Key limitation	GRADE Quality
Acute / hepatic migratory phase	Exposure history, eosinophilia, serology, ultrasound/CT, clinical syndrome	Eggs often absent before adult flukes reach bile ducts	Moderate
Chronic / biliary phase	Repeated stool microscopy, rapid sedimentation, Kato-Katz, duodenal/biliary aspirate, serology	Intermittent egg shedding; species eggs may be morphologically indistinguishable	Moderate
Complicated biliary disease	Imaging, ERCP where indicated, stool/serology support	Requires specialist care	Low to Moderate

CDC DPDx states that eggs can be detected during the chronic adult phase in stool or duodenal/biliary material, while serologic testing can support diagnosis; CDC's FhSAP2 immunoblot has reported chronic infection sensitivity $\geq 94\%$ and specificity $\geq 98\%$, but has not been validated for acute infection [CDC DPDx](#).

C. Treatment approach

Patient / setting	Recommended treatment	Follow-up	GRADE Quality	Strength
Confirmed or strongly suspected fascioliasis	Triclabendazole 10 mg/kg orally once , WHO standard regimen	Reassess symptoms and, in chronic infection, repeat stool testing	Moderate	Strong
Treatment failure after standard WHO dose	Triclabendazole 20 mg/kg total , divided into two doses 12–24 hours apart	Confirm adherence, food intake, diagnosis, reinfection risk, and persistent eggs/antigen	Low to Moderate	Conditional
U.S. patients ≥ 6 years	CDC/FDA regimen: two 10 mg/kg doses 12 hours apart with food	Clinical and parasitologic follow-up	Moderate	Strong
Bile duct obstruction or severe complications	Triclabendazole plus specialist evaluation; surgery or ERCP rarely needed	Monitor for obstruction, cholangitis, pancreatitis	Low	Conditional
Children <6 years or pregnancy	Expert consultation	Individualized risk-benefit decision	Very Low	Conditional

WHO identifies triclabendazole as first-line therapy [WHO Q&A](#), while CDC recommends two 10 mg/kg oral doses 12 hours apart with food for patients ≥ 6 years [CDC Clinical Overview](#).

Final Dynamic Guideline Position as of 29 April 2026

1. Triclabendazole remains the WHO-based first-line treatment for fascioliasis.

The WHO standard is **10 mg/kg orally once**, with **20 mg/kg total in two divided doses** for treatment failure [WHO recommendations](#).

2. Diagnosis should be multimodal.

Stool egg detection confirms chronic disease, but acute fascioliasis often requires exposure history, eosinophilia, serology, and imaging because eggs may be absent early [CDC DPDx](#), [PMC review](#).

3. Treatment failure and resistance require active surveillance.

Human treatment